Supplementary Table 1. Search Strategies for database.

PubMed Results:20

| Search number | Search Details |
|---|---|
| #1 | (("receptors, melatonin"[MeSH Terms] OR ("receptors"[All Fields] AND "melatonin"[All Fields]) OR "melatonin receptors"[All Fields] OR ("melatonin"[All Fields] AND "receptor"[All Fields]) OR "melatonin receptor"[All Fields]) AND ("agonist"[All Fields] OR "agonist s"[All Fields] OR "agonistic"[All Fields] OR "agonistically"[All Fields] OR "agonistics"[All Fields] OR "agonists"[MeSH Subheading] OR "agonists"[All Fields])) OR ("melatonin"[MeSH Terms] OR "melatonin"[All Fields] OR "n acetyl 5 methoxytryptamine"[All Fields]) OR ("melatonin"[MeSH Terms] OR "melatonin"[All Fields] OR "melatonin s"[All Fields] OR "melatonine"[All Fields] OR "melatonins"[All Fields]) OR ("ramelteon"[Supplementary Concept] OR "ramelteon"[All Fields]) OR ("tasimelteon"[Supplementary Concept] OR "tasimelteon"[All Fields]) OR ("s 20098"[Supplementary Concept] OR "s 20098"[All Fields] OR "agomelatine"[All Fields] OR "agomelatin"[All Fields] OR "agomelatine s"[All Fields]) OR "Melaxen"[All Fields] OR "Melatonergics"[All Fields] |
| #2 | "thoracic surgery"[MeSH Terms] OR ("thoracic"[All Fields] AND "surgery"[All Fields]) OR "thoracic surgery"[All Fields] OR ("cardiac"[All Fields] AND "surgery"[All Fields]) OR "cardiac surgery"[All Fields] OR "cardiac surgical procedures"[MeSH Terms] OR ("cardiac"[All Fields] AND "surgical"[All Fields] AND "procedures"[All Fields]) OR "cardiac surgical procedures"[All Fields] OR ("cardiac"[All Fields] AND "surgery"[All Fields]) OR ("cardiothoracic"[All Fields] AND ("surgery"[MeSH Subheading] OR "surgery"[All Fields] OR "surgical procedures, operative"[MeSH Terms] OR ("surgical"[All Fields] AND "procedures"[All Fields] AND "operative"[All Fields]) OR "operative surgical procedures"[All Fields] OR "general surgery"[MeSH Terms] OR ("general"[All Fields] AND "surgery"[All Fields]) OR "general surgery"[All Fields] OR "surgery s"[All Fields] OR "surgerys"[All Fields] OR "surgeries"[All Fields])) OR ("coronary artery bypass"[MeSH Terms] OR ("coronary"[All Fields] AND "artery"[All Fields] AND "bypass"[All Fields]) OR "coronary artery bypass"[All Fields]) OR (("surgical procedures, operative"[MeSH Terms] OR ("surgical"[All Fields] AND "procedures"[All Fields] AND "operative"[All Fields]) OR "operative surgical procedures"[All Fields] OR "surgical"[All Fields] OR "surgically"[All Fields] OR "surgicals"[All Fields]) AND ("coronaries"[All Fields] OR "heart"[MeSH Terms] OR "heart"[All Fields] OR "coronary"[All Fields]) AND ("revascularisation"[All Fields] OR "revascularisations"[All Fields] OR "revascularise"[All Fields] OR "revascularised"[All Fields] OR "revascularising"[All Fields] OR "revascularization"[All Fields] OR "revascularizations"[All Fields] OR "revascularize"[All Fields] OR "revascularized"[All Fields] OR "revascularizes"[All Fields] OR "revascularizing"[All Fields])) OR (("valve"[All Fields] OR "valve s"[All Fields] OR "valved"[All Fields] OR "valves"[All Fields] OR "valving"[All Fields]) AND ("surgery"[MeSH Subheading] OR "surgery"[All Fields] OR "surgical procedures, operative"[MeSH Terms] OR ("surgical"[All Fields] AND "procedures"[All Fields] AND "operative"[All Fields]) OR "operative surgical procedures"[All Fields] OR "general surgery"[MeSH Terms] OR ("general"[All Fields] AND "surgery"[All Fields]) OR "general surgery"[All Fields] OR "surgery s"[All Fields] OR "surgerys"[All Fields] OR "surgeries"[All Fields])) OR (("valve"[All Fields] OR "valve s"[All Fields] OR "valved"[All Fields] OR "valves"[All Fields] OR "valving"[All Fields]) AND ("replace"[All Fields] OR "replaceable"[All Fields] OR "replaced"[All Fields] OR "replaces"[All Fields] OR "replacing"[All Fields] OR "replacment"[All Fields] OR "replantation"[MeSH Terms] OR "replantation"[All Fields] OR "replacement"[All Fields] OR "replacements"[All Fields])) |

| | |
|---|---|
| #3 | "delirium"[MeSH Terms] OR "delirium"[All Fields] OR "deliriums"[All Fields] OR ("confusability"[All Fields] OR "confusable"[All Fields] OR "confuse"[All Fields] OR "confuses"[All Fields] OR "confusing"[All Fields] OR "confusion"[MeSH Terms] OR "confusion"[All Fields] OR "confused"[All Fields] OR "confusions"[All Fields]) OR ("confusional"[All Fields] AND ("syndrom"[All Fields] OR "syndromal"[All Fields] OR "syndromally"[All Fields] OR "syndrome"[MeSH Terms] OR "syndrome"[All Fields] OR "syndromes"[All Fields] OR "syndrome s"[All Fields] OR "syndromic"[All Fields] OR "syndroms"[All Fields])) OR (("postoperative period"[MeSH Terms] OR ("postoperative"[All Fields] AND "period"[All Fields]) OR "postoperative period"[All Fields] OR "postop"[All Fields] OR "postoperative"[All Fields] OR "postoperatively"[All Fields] OR "postoperatives"[All Fields]) AND ("delirium"[MeSH Terms] OR "delirium"[All Fields] OR "deliriums"[All Fields])) OR ("cognition disorders"[MeSH Terms] OR ("cognition"[All Fields] AND "disorders"[All Fields]) OR "cognition disorders"[All Fields] OR ("cognitive"[All Fields] AND "disorder"[All Fields]) OR "cognitive disorder"[All Fields]) |
| #4 | (#1) AND (#2)) AND (#3) |

Cochrane Library Results:17

| Search number | Search Details |
|---|---|
| 1 | (melatonin AND receptor AND agonist OR n-acetyl-5-methoxytryptamine OR melatonin OR ramelteon OR tasimelteon OR agomelatine OR melaxen OR melatonergics) |
| 2 | (cardiac AND surgery OR (cardiothoracic AND surgery) OR (coronary AND artery AND bypass) OR (surgical AND coronary AND revascularization) OR (valve AND surgery) OR (valve AND replacement) |
| 3 | (delirium OR confusion OR (confusional AND syndrome) OR (postoperative AND delirium) OR (valve AND surgerycognitive AND disorder) |
| 4 | 1 AND 2 AND 3 |

Web of Science Results:27

| Search number | Search Details |
|---|---|
| #1 | ALL FIELDS: (melatonin receptor agonist) OR ALL FIELDS: (N-acetyl-5-methoxytryptamine) OR ALL FIELDS: (melatonin) OR ALL FIELDS: (ramelteon) OR ALL FIELDS: (tasimelteon) OR ALL FIELDS: (agomelatine) OR ALL FIELDS: (Melaxen) OR ALL FIELDS: (Melatonergics)<br>Indexes=SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, BKCI-S, BKCI-SSH, ESCI, CCR-EXPANDED, IC Timespan=All years |
| #2 | ALL FIELDS: (cardiac surgery) OR ALL FIELDS: (cardiothoracic surgery) OR ALL FIELDS: (coronary artery bypass) OR ALL FIELDS: (surgical coronary revascularization) OR ALL FIELDS: (valve surgery) OR ALL FIELDS: (valve replacement)<br>Indexes=SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, BKCI-S, BKCI-SSH, ESCI, CCR-EXPANDED, IC Timespan=All years |
| #3 | ALL FIELDS: (delirium) OR ALL FIELDS: (confusion) OR ALL FIELDS: (confusional syndrome) OR ALL FIELDS: (postoperative delirium) OR ALL FIELDS: (cognitive disorder)<br>Indexes=SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, BKCI-S, BKCI-SSH, ESCI, CCR-EXPANDED, IC Timespan=All years |
| #4 | #3 AND #2 AND #1<br>Indexes=SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, BKCI-S, BKCI-SSH, ESCI, CCR-EXPANDED, IC Timespan=All years |

Embase Results:30

| Search number | Search Details |
|---|---|
| 1 | (melatonin AND receptor AND agonist OR n-acetyl-5-methoxytryptamine OR melatonin OR ramelteon OR tasimelteon OR agomelatine OR melaxen OR melatonergics) |
| 2 | (cardiac AND surgery OR (cardiothoracic AND surgery) OR (coronary AND artery AND bypass) OR (surgical AND coronary AND revascularization) OR (valve AND surgery) OR (valve AND replacement) |
| 3 | (delirium OR confusion OR (confusional AND syndrome) OR (postoperative AND delirium) OR (valve AND surgerycognitive AND disorder) |
| 4 | 1 AND 2 AND 3<br>Mapped terms n/a |

Ebscohost Results:118

| Search number | Search Details |
|---|---|
| S1 | (TX+delirium+OR+TX+confusion+OR+TX+confusional+syndrome+OR+TX+postoperative+delirium+OR+TX+cognitive+disorder) |
| S2 | (TX+melatonin+receptor+agonist)+OR+(TX+N%E2%80%90acetyl%E2%80%905%E2%80%90methoxytryptamine)+OR+(TX+melatonin)+OR+(TX+ramelteon)+OR+(TX+tasimelteon)+OR+(TX+agomelatine)+OR+(TX+Melaxen)+OR+(TX+Melatonergics) |
| S3 | (TX+delirium)+OR+(TX+confusion)+OR+(TX+confusional+syndrome)+OR+(TX+postoperative+deliriu m)+OR+(TX+cognitive+disorder) |
| S4 | S1 AND S2 AND S3 (lang=zh-cn&type=0&searchMode=Standard&site=ehost-live) |

Supplementary Table 2. Risk of bias assessment for RCTs

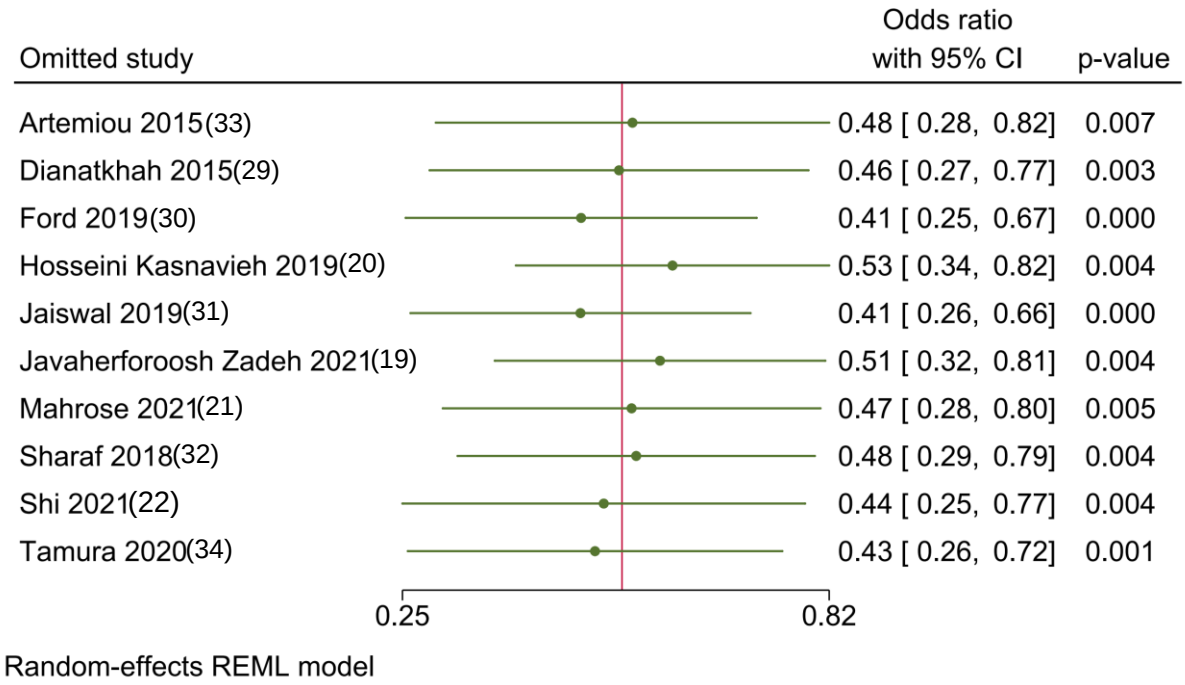
| Author, Year | Randomization process | Deviations from intended interventions | Missing outcome data | Measurement of the outcome | Selection of the reported result | Overall |
|---|---|---|---|---|---|---|
| Dianatkhah 2015(29) | + | ? | + | ? | + | ? |
| Ford 2019(30) | + | + | + | + | + | + |
| Hosseini Kasnavieh 2019(20) | + | + | + | + | + | + |
| Jaiswal 2019(31) | + | + | + | + | + | + |
| Javaherforoosh Zadeh 2021(19) | + | + | + | + | + | + |
| Mahrose 2021(21) | ? | + | + | + | + | ? |
| Sharaf 2018(32) | + | + | ? | + | + | + |
| Shi 2021(22) | + | + | + | + | + | + |

+ Low risk
? Some concerns
? High risk

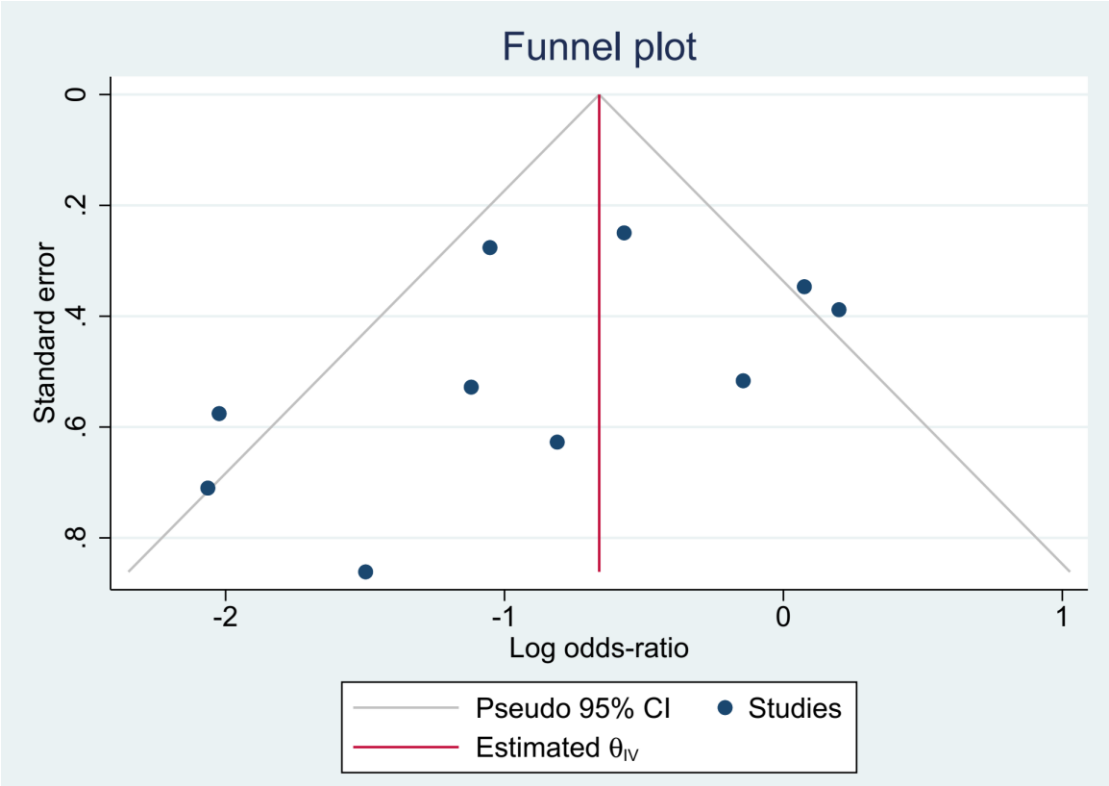
Supplementary Table 3. Risk of bias assessment for non-RCTs

| study | Study type | Selection | Comparability | Outcome or Exposure | Total score |
|---|---|---|---|---|---|
| Artemiou 2015(33) | prospective cohort study | **** | * | *** | 8 |
| Tamura 2020(34) | retrospective cohort study | **** | ** | *** | 9 |

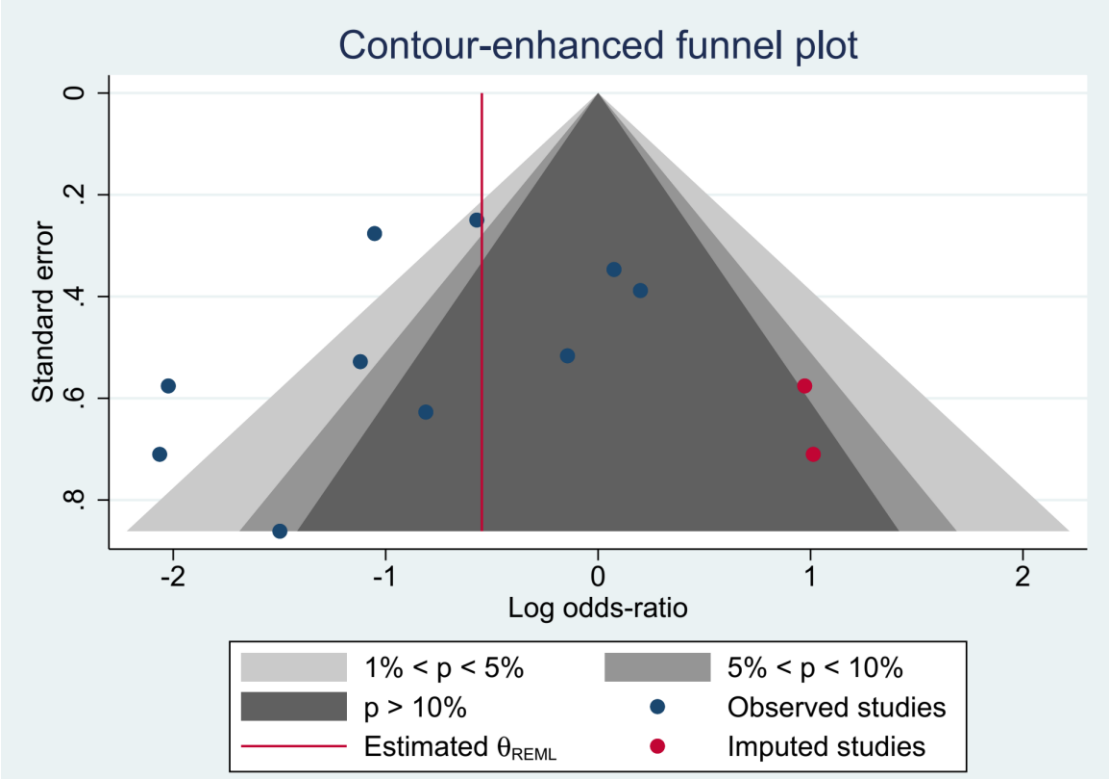
Supplementary Figure 1. Forest plot for leave-one-out meta-analysis.



Supplementary Figure 2. Funnel plot for postoperative delirium incidence.




Supplementary Figure 3. Trim-and fill meta-analysis.



```
. meta trimfill, estimator(linear) eform

  Effect-size label:  Log Odds-Ratio
        Effect size:  _meta_es
          Std. Err.:  _meta_se

Nonparametric trim-and-fill analysis of publication bias
Linear estimator, imputing on the right

Iteration                             Number of studies =     12
  Model: Random-effects                       observed =     10
 Method: REML                                 imputed =      2

Pooling
  Model: Random-effects
 Method: REML
```

| Studies | Odds Ratio | [95% Conf. | Interval] |
|---|---|---|---|
| Observed | 0.461 | 0.287 | 0.742 |
| Observed + Imputed | 0.579 | 0.340 | 0.983 |